

Applied Machine Learning

Evaluation of Scorers and Rankers



UNIVERSITY OF
GOTHENBURG

CHALMERS

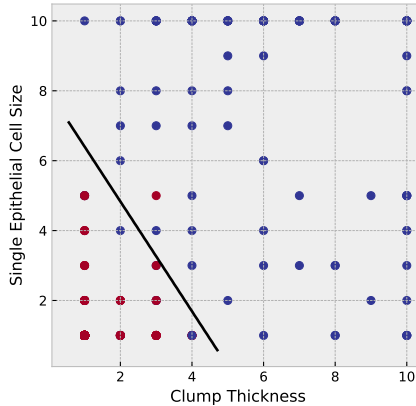
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what if it's more important to find the malignant tumors?

- ▶ how can we adjust our decisions?

breast cancer example

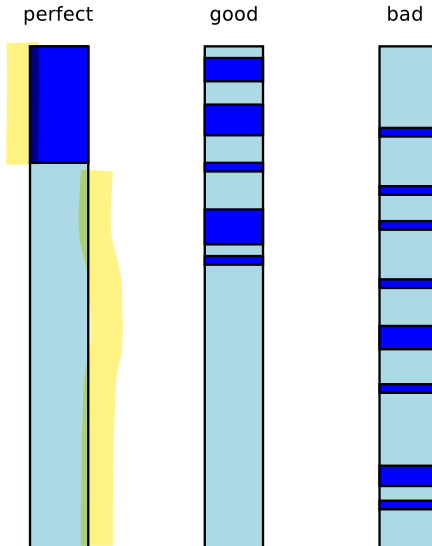


when we evaluate, can we abstract away from the threshold?

ranking and scoring systems

- ▶ many classifiers output some sort of **confidence score**
 - ▶ `decision_function`, `predict_proba`, ...
- ▶ the methods we discuss now are more or less the same as those used to **evaluate search engine** (or more generally, **ranking systems**)

intuition: evaluating rankers



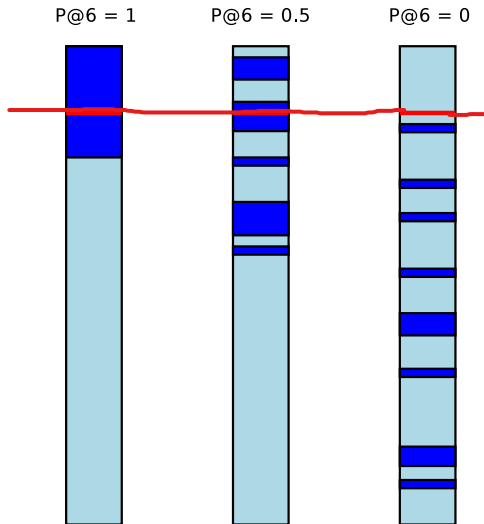
evaluating a search engine: quality of the first page

- ▶ when I use a search engine, how good are the results on the first page?
- ▶ **precision at k**

$$P@k = \frac{\text{nbr relevant items in top } k}{k}$$

The screenshot shows a Google search interface with the query "applied machine learning gu chalmers exam". The search results are displayed in a list format. The first result is titled "DIT865 V18 Applied Machine learning - GUL" and includes a link to the course homepage. The second result is titled "DIT865 V18 Applied Machine learning - GUL" and includes a link to the course homepage. The third result is titled "Course PM - DIT865 V18 Applied Machine learning (GUL)" and includes a link to the course homepage. The fourth result is titled "Algorithms for Machine Learning & Inference | CSE LAB" and includes a link to the course homepage. The fifth result is titled "Search course | Chalmers studentportal" and includes a link to the student portal. The sixth result is titled "Course-PM TIN093/DIT600 Algorithms" and includes a link to the course homepage.

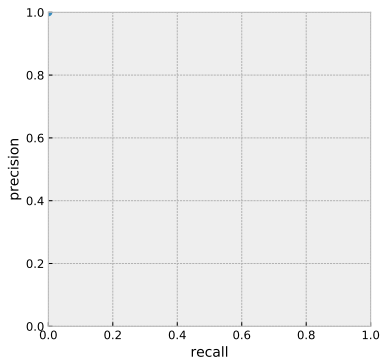
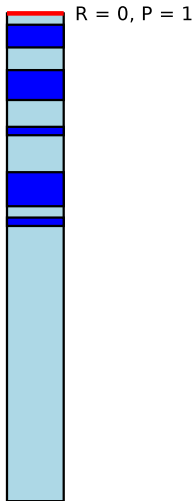
evaluating a search engine (another example)



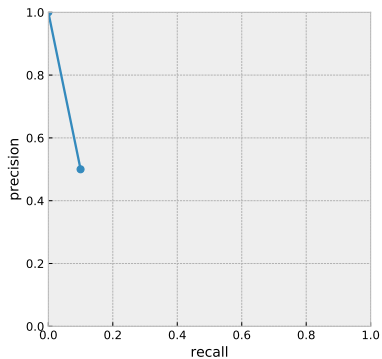
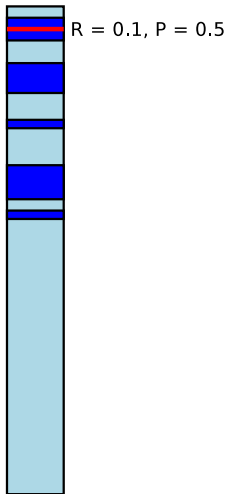
precision / recall curve

- ▶ is there a way to evaluate ranker or scorer that does not depend on the threshold?
- ▶ in a **precision/recall curve**, we compute the precision for different recall levels
 - ▶ that is, for different thresholds
- ▶ we visualize the relationship using a plot
 - ▶ good = curve is near top right

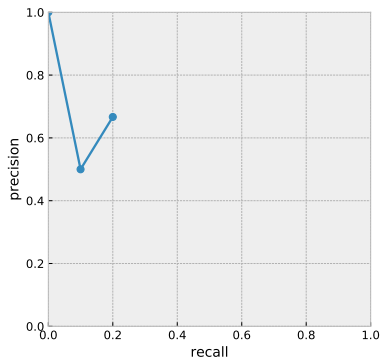
P / R curve (example)



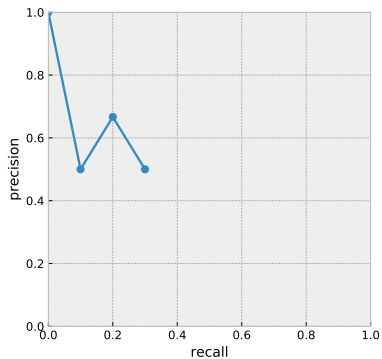
P / R curve (example)



P / R curve (example)



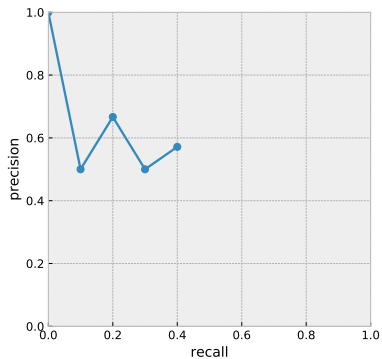
P / R curve (example)



P / R curve (example)



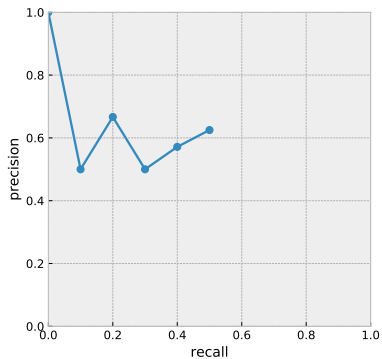
$R = 0.4, P = 0.57$



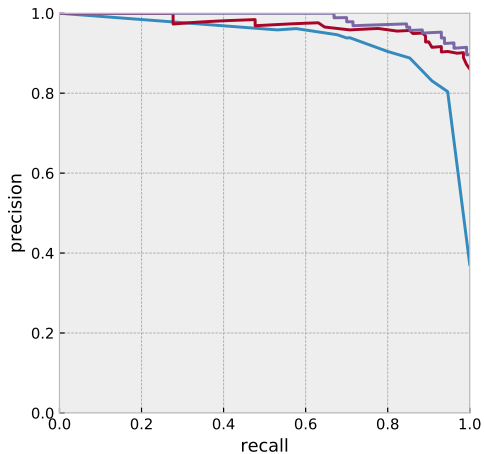
P / R curve (example)



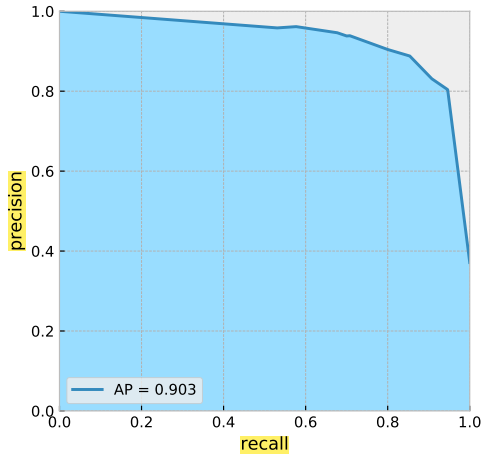
$R = 0.5, P = 0.63$



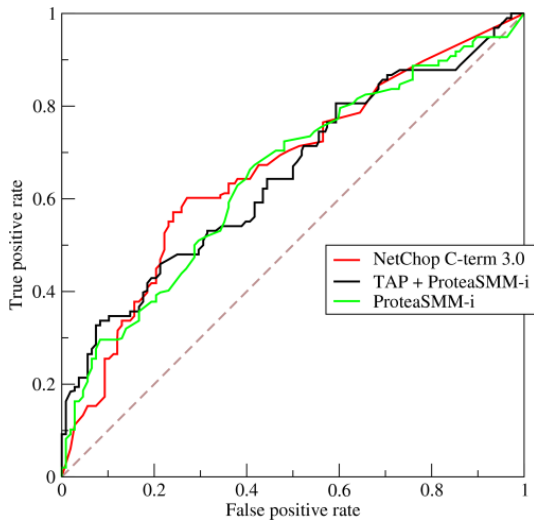
comparing P/R curves



average precision

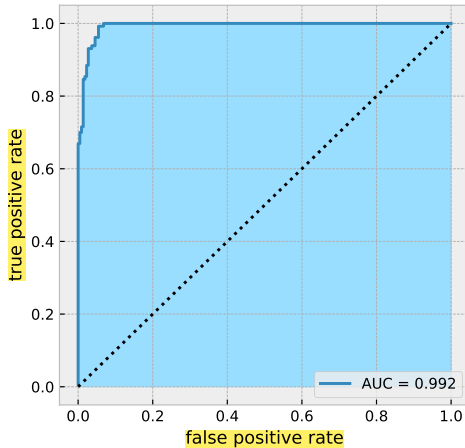


ROC curve (Receiver Operating Characteristic)



[[source](#)]

Area Under Curve score



summary: evaluating scoring and ranking systems

- ▶ regular classification metrics typically depend on a **threshold**
- ▶ we may want to consider the system's behavior for different values of the threshold
 - ▶ drawing a **curve** to explore the tradeoff
 - ▶ computing an **aggregate value** over all thresholds (**AP**, **AUC**)
- ▶ methods are similar to those used to evaluate **search engines**